

Chapter 4: Categorical and Mixed-type data (5 hrs)

1. **Classification: Decision Trees and the CART algorithm**
2. **Regression: Logistic Regression**
3. **Variable Importance: Permutation tests, Partial dependence plots**
4. **Clustering: K-means and distance metrics for mixed-type data**

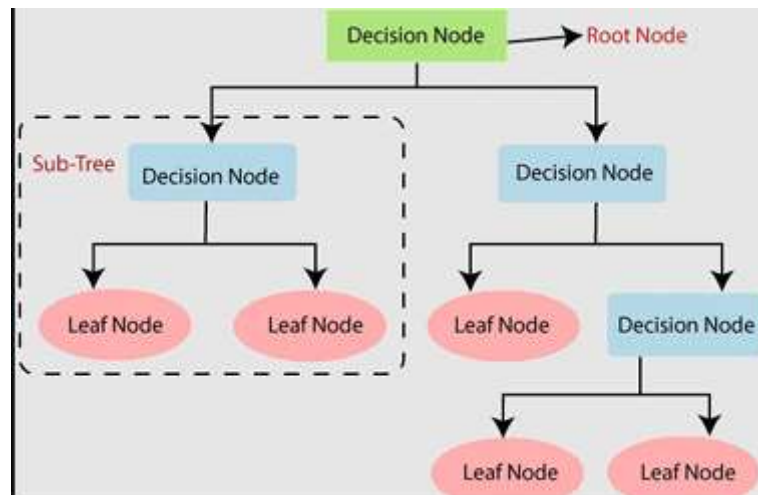
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4. Categorical and Mixed-type Data

1. Classification: Decision Trees and the CART algorithm

What is a Decision Tree?

A decision tree is a type of supervised machine learning algorithm used for both classification and regression tasks. It works by splitting the dataset into smaller and smaller subsets based on certain conditions, forming a tree-like structure. The goal is to find the best splits that help separate the data points clearly based on the target variable.

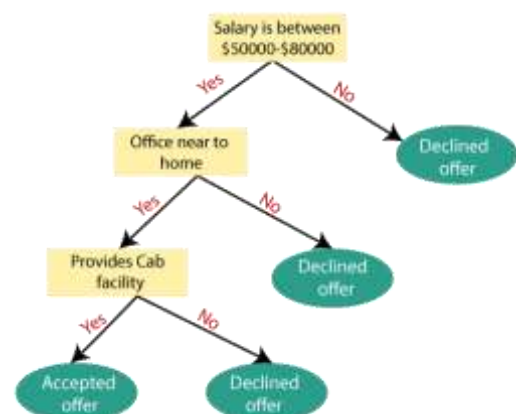


It consists of following components:

- Root Node:** The topmost node. It represents the entire dataset, and the first decision (split) is made here.
- Internal/Decision Nodes:** These are the points where the dataset is split based on a feature.
- Branches:** These represent the outcome of a split (e.g., Yes or No, $\leq$ threshold or $>$ threshold).
- Leaf/Terminal Nodes:** These are the final outcomes (classes or predicted values)

How Does It Work?

- In a decision tree, for predicting the class of the given dataset, the algorithm starts from the root node of the tree.
- This algorithm compares the values of root attribute with the record (real dataset) attribute and, based on the comparison,



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follows the branch and jumps to the next node.

- For the next node, the algorithm again compares the attribute value with the other sub-nodes and move further. It continues the process until it reaches the leaf node of the tree.

Pros and Cons of Decision Trees

Advantages

- Easy to understand and visualize
- No need for feature scaling or normalization
- Works with both numerical and categorical data

Disadvantages

- Prone to overfitting
- Can be unstable — small data changes may produce different trees
- Biased toward features with many levels/categories

Real-world Applications

- Medical Diagnosis (e.g., classifying tumors as benign or malignant)
- Credit Scoring
- Customer Churn Prediction
- Fraud Detection

Decision Tree Construction:

a) *Tree Construction*

- Select the best feature to split the data at each node.
- Use a metric to measure the quality of splits (e.g., Gini Index, Information Gain).

b) *Recursive Splitting*

- Subdivide data at each node until stopping criteria are met (e.g., maximum depth, minimum samples per node).

c) *Tree Pruning*

- Reduce overfitting by removing unnecessary branches (post-pruning or pre-pruning).

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- *Pruning is a process of deleting the unnecessary nodes from a tree in order to get the optimal decision tree.*

Attribute Selection Measures:

The decision tree algorithm chooses the best feature and split point using a criterion such as:

- a. Information Gain (used in ID3)
- b. Gain Ratio (used in C4.5)
- c. Gini Impurity (used in CART)

a) Decision Tree Construction with ID3 Algorithm (with Information Gain):

- Information gain is used by ID3 algorithm.
- Information gain is the measurement of changes in entropy after the segmentation of a dataset based on an attribute.
- It calculates how much information a feature provides us about a class.
- It can be calculated using the below formula:

$$\text{Information Gain} = \text{Entropy}(S) - [(\text{Weighted Avg}) * \text{Entropy}(\text{each feature})]$$

- The entropy is a measure of the uncertainty associated with a random variable
- For given data partition S:

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

- Conditional Entropy is given by:

$$E(T, X) = \sum_{c \in X} P(c) E(c)$$

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Numerical:

Given the following dataset:

Day	Temp	Humidity	Wind	Play
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D3	Cool	High	Weak	Yes
D4	Cool	High	Strong	Yes
D5	Hot	Normal	Weak	Yes
D6	Cool	Normal	Weak	Yes
D7	Cool	Normal	Strong	No

1. What is the entropy of this collection of training examples with respect to the target function *Play*?
2. What is the information gain of *Temp*, *Humidity* and *Wind* relative to these training examples?
3. Draw decision tree for the given dataset using ID3 algorithm

Solution:

Step 1: Computation of entropy of each attributes

$$\begin{aligned}\text{Entropy (Play)} &= \text{Entropy (4,3)} \\ &= -\left(\frac{4}{7} \log_2 \frac{4}{7}\right) - \left(\frac{3}{7} \log_2 \frac{3}{7}\right) \\ &= -(-0.461) - (-0.523) \\ &= 0.985\end{aligned}$$

Now,

$$\begin{aligned}\text{Entropy (Play, Temp)} &= P(\text{Hot}) * E(1,2) + P(\text{Cool}) * E(3,1) \\ &= \frac{3}{7} * \left[-\left(\frac{1}{3} \log_2 \frac{1}{3}\right) - \left(\frac{2}{3} \log_2 \frac{2}{3}\right)\right] + \frac{4}{7} * \left[-\left(\frac{3}{4} \log_2 \frac{3}{4}\right) - \left(\frac{1}{4} \log_2 \frac{1}{4}\right)\right] \\ &= 0.3935 + 0.4636 \\ &= 0.8570\end{aligned}$$
$$\begin{aligned}\text{Entropy (Play, Humidity)} &= P(\text{High}) * E(2,2) + P(\text{Normal}) * E(2,1) \\ &= \frac{4}{7} * \left[-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4}\right] + \frac{3}{7} * \left[-\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}\right] \\ &= 0.5714 + 0.3935 \\ &= 0.9649\end{aligned}$$

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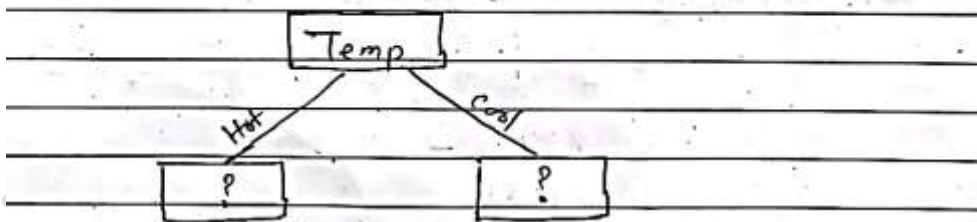
$$\begin{aligned}
 \text{Entropy}(\text{Play}, \text{Wind}) &= P(\text{Strong}) \cdot E(1, 2) + P(\text{Weak}) \cdot E(3, 1) \\
 &= \frac{3}{7} \cdot \left[-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \right] + \frac{4}{7} \cdot \left[-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right] \\
 &= 0.3935 + 0.4636 \\
 &= 0.8571
 \end{aligned}$$

Computation of information gain:

$$\begin{aligned}
 \text{Gain}(\text{Play}, \text{Temp}) &= \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Temp}) \\
 &= 0.985 - 0.8570 \\
 &= 0.128 \\
 \text{Gain}(\text{Play}, \text{Humidity}) &= \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Humidity}) \\
 &= 0.985 - 0.9649 \\
 &= 0.0201 \\
 \text{Gain}(\text{Play}, \text{Wind}) &= \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Wind}) \\
 &= 0.985 - 0.8571 \\
 &= 0.1279
 \end{aligned}$$

Construction of partial decision tree with root node.

Since, the $\text{Gain}(\text{Play}, \text{Temp})$ has the highest information gain, we place this attribute (Temp) at the root node.



Step 2: Now again we continue to compute the entropy of each attribute (*Humidity*, *wind*) with reference to the data value of *Temp* attribute. The data values are *Hot* and *Cool*.

- The new table with reference to *Hot* is:

Day	Humidity	Wind	Play
D1	High	Weak	No
D2	High	Strong	No
D5	Normal	Weak	Yes

- Now we compute Entropy and Information gain as in Step 1.

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$$\begin{aligned}
 \text{Entropy}(\text{Play}_{\text{hot}}) &= \text{Entropy}(1, 2) \\
 &= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \\
 &= 0.9183
 \end{aligned}$$

$$\begin{aligned}
 \text{Entropy}(\text{Play}_{\text{hot}}, \text{humidity}) &= P(\text{high}) * E(0, 2) + P(\text{Normal}) * E(1, 0) \\
 &= \frac{2}{3} \left[-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \right] + \frac{1}{3} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right] \\
 &= 0 + 0 \\
 &= 0
 \end{aligned}$$

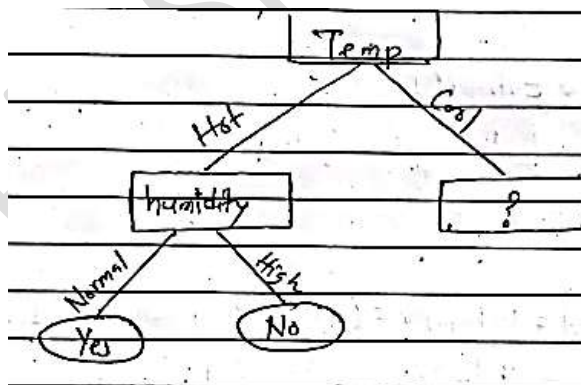
$$\begin{aligned}
 \text{Entropy}(\text{Play}_{\text{hot}}, \text{Wind}) &= P(\text{Strong}) * E(0, 1) + P(\text{Weak}) * E(1, 1) \\
 &= \frac{1}{3} \left[-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right] + \frac{2}{3} \left[-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right] \\
 &= 0 + 0.66 \\
 &= 0.66
 \end{aligned}$$

Now,

$$\begin{aligned}
 \text{Gain}(\text{Play}_{\text{hot}}, \text{humidity}) &= \text{Entropy}(\text{Play}_{\text{hot}}) - \text{Entropy}(\text{Play}_{\text{hot}}, \text{humidity}) \\
 &= 0.9183 - 0 \\
 &= 0.9183
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain}(\text{Play}_{\text{hot}}, \text{wind}) &= \text{Entropy}(\text{Play}_{\text{hot}}) - \text{Entropy}(\text{Play}_{\text{hot}}, \text{Wind}) \\
 &= 0.9183 - 0.66 \\
 &= 0.2583
 \end{aligned}$$

- Since the information gain of humidity is high we take it as the parent node. Our partial decision tree is:

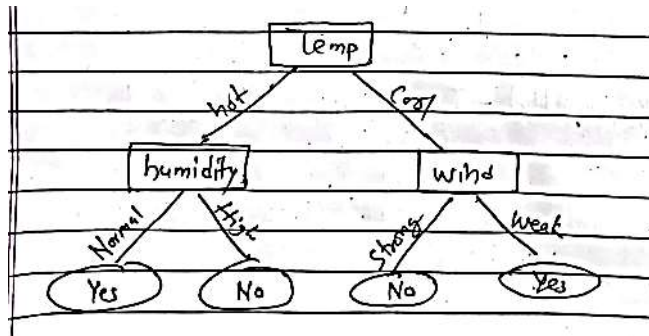


Step 3: Similar to step 2, we need to compute the entropy and information gain for (PlayCool) with reference to *wind* and *humidity* attribute with below table.

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Day	Humidity	Wind	Play
D3	High	Weak	Yes
D4	High	Strong	Yes
D6	Normal	Weak	Yes
D7	Normal	Strong	No

But, since the remaining is *wind* attribute only. So, we can directly draw the final decision tree as below



b) Decision Tree Construction with C4.5 Algorithm (with Gain ratio):

- Gain Ratio is an alternative to Information Gain that is used to select the attribute for splitting in a decision tree.
- Used by C4.5 (the successor of ID3)
- It is used to **overcome the problem of bias** towards the attribute with many outcomes.
- Gain Ratio **normalizes the Information Gain with respect to the total entropy** of all splits based on values of an attribute
- It introduces a normalizing term called **Intrinsic Information**.
- Gain ratio is given by:

$$\text{GainRatio} = \frac{\text{Information Gain}}{\text{Intrinsic Information}}$$

- Where:

$$\text{Intrinsic Information} = - \sum_v \frac{|S_v|}{|S|} \cdot \log_2 \left(\frac{|S_v|}{|S|} \right)$$

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- Where:
 - $|S|$ = total number of samples before the split.
 - $|S_v|$ = number of samples in subset v after the split.
 - v iterates over all possible values of the splitting attribute.

Example:

Day	Temp	Humidity	Wind	Play
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D3	Cool	High	Weak	Yes
D4	Cool	High	Strong	Yes
D5	Hot	Normal	Weak	Yes
D6	Cool	Normal	Weak	Yes
D7	Cool	Normal	Strong	No

Here, Intrinsic Information of Temp is :

$$= - \left(\frac{4}{7} \cdot \log_2 \frac{4}{7} + \frac{3}{7} \cdot \log_2 \frac{3}{7} \right)$$
$$= 0.985$$

$$\text{Gain ratio}_{\text{Temp}} = 0.128 / 0.9851 = 0.01299$$

c) Decision Tree Construction with CART Algorithm (with Gini Index):

- Gini index is a **measure of impurity or purity used** while creating a decision tree in the CART (Classification and Regression Tree) algorithm.
- **An attribute with the low Gini index should be preferred** as compared to the high Gini index.
- It only creates binary splits, and the CART algorithm uses the Gini index to create binary splits.
- Gini index can be calculated using the below formula:

$$\text{Gini Index} = 1 - \sum_j p_j^2$$

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Numerical:

Compute the Gini Index for overall collection of the training examples, and construct a decision tree for below training dataset.

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

Step 1: Computation of Gini Index for target class attribute

$$\begin{aligned} \text{Gini Index for Decision,} \\ \text{Gini}(S) &= 1 - \left[\left(\frac{6}{10}\right)^2 + \left(\frac{2}{10}\right)^2 + \left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right)^2 \right] \\ &= 0.58 \end{aligned}$$

Step 2: Computation of Gini Index for other attributes as well.

$$\begin{aligned} \text{Computation of Gini Index for 'Money'} \\ \text{Gini}(\text{Money}_{\text{poor}}) &= 1 - \left[\left(\frac{9}{9}\right)^2 \right] \\ &= 0 \\ \text{Gini}(\text{Money}_{\text{rich}}) &= 1 - \left[\left(\frac{2}{7}\right)^2 + \left(\frac{3}{7}\right)^2 + \left(\frac{1}{7}\right)^2 + \left(\frac{1}{7}\right)^2 \right] \\ &= 0.694 \\ \text{Weighted average (Money)} \\ \text{Gini}(\text{Money}) &= 0 \times \frac{9}{10} + 0.694 \times \frac{7}{10} \\ &= 0.486 \end{aligned}$$

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Computation of Gini Index for "Parents" attribute.

$$\text{Gini}(\text{Parent}_{\text{yes}}) = 1 - \left[\left(\frac{5}{5} \right)^2 \right] \\ = 0$$

$$\text{Gini}(\text{Parent}_{\text{No}}) = 1 - \left[\left(\frac{2}{5} \right)^2 + \left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^2 \right] \\ = 0.72$$

Now, Weighted average (Parents)

$$\text{Gini}(\text{Parent}) = 0 * \left(\frac{5}{10} \right) + 0.72 * \left(\frac{5}{10} \right) \\ = 0.36$$

Computation of Gini Index for "Weather" attribute.

$$\text{Gini}(\text{Weather}_{\text{sunny}}) = 1 - \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] \\ = 0.444$$

$$\text{Gini}(\text{Weather}_{\text{rainy}}) = 1 - \left[\left(\frac{2}{2} \right)^2 + \left(\frac{1}{3} \right)^2 \right] \\ = 0.444$$

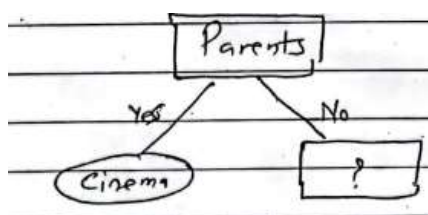
$$\text{Gini}(\text{Weather}_{\text{windy}}) = 1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] \\ = 0.375$$

Weighted average (Weather)

$$\text{Gini}(\text{Weather}) = 0.444 * \frac{3}{10} + 0.444 * \frac{2}{10} + 0.375 * \frac{4}{10} \\ = 0.416$$

Since the Gini Index of **Parent** is the smallest, we put this attribute at the root node.

Our partial decision tree is:



Now our remaining table with **Parents** = **No** is:

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Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Rainy	No	Rich	Stay In
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W10	Sunny	No	Rich	Tennis

Step 3: Computation of Gini Index for **Weather** and **Money** with **Parents=No**

Computation of Gini Index for attribute **Weather**:

$$\begin{aligned} \text{Gini}(\text{Weather}_{\text{sunny}}) &= 1 - \left[\left(\frac{2}{2} \right)^2 \right] \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Gini}(\text{Weather}_{\text{rainy}}) &= 1 - \left[\left(\frac{1}{2} \right)^2 \right] \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Gini}(\text{Weather}_{\text{windy}}) &= 1 - \left[\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \right] \\ &= 0.5 \end{aligned}$$

Weighted average Gini

$$\begin{aligned} \text{Gini}(\text{Weather}) &= 0 * \frac{2}{5} + 0 * \frac{1}{5} + 0.5 * \frac{2}{5} \\ &= 0.2 \end{aligned}$$

Computation of Gini Index for attribute **Money**:

$$\begin{aligned} \text{Gini}(\text{Money}_{\text{rich}}) &= 1 - \left[\left(\frac{2}{4} \right)^2 + \left(\frac{1}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] \\ &= 0.625 \end{aligned}$$

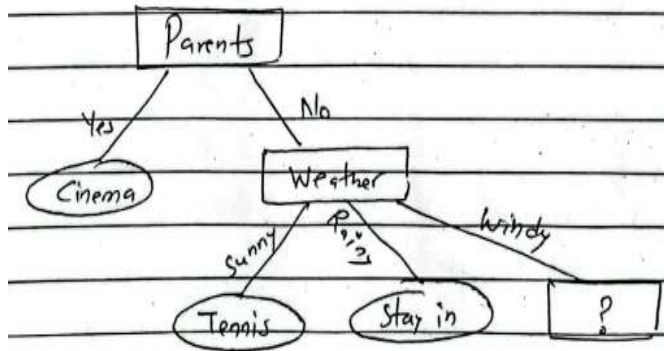
$$\begin{aligned} \text{Gini}(\text{Money}_{\text{poor}}) &= 1 - \left[\left(\frac{1}{1} \right)^2 \right] \\ &= 0 \end{aligned}$$

Weighted average Gini

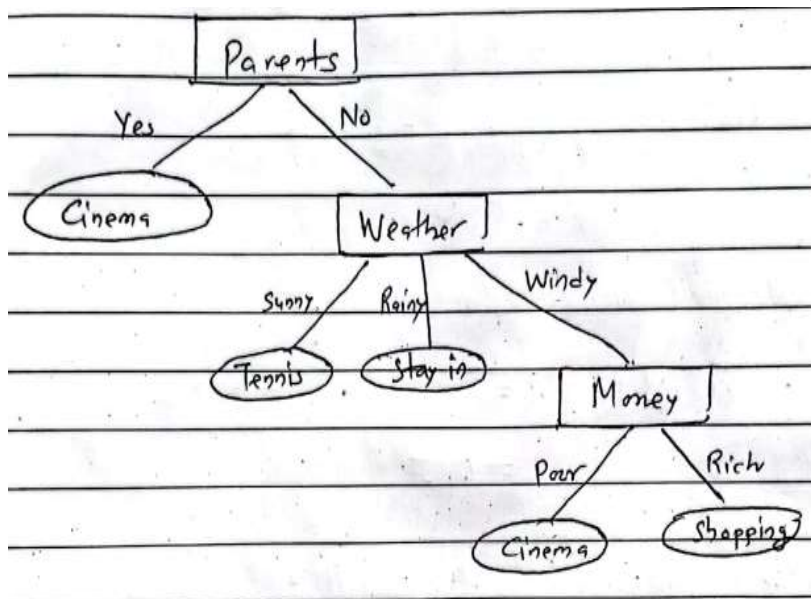
$$\begin{aligned} \text{Gini}(\text{Money}) &= 0.625 * \frac{4}{5} + 0 * \frac{1}{5} \\ &= 0.5 \end{aligned}$$

Here, **weather** has the smallest Gini Index. So, it is selected as the next node in the decision tree as shown below:

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Definitely, the next remaining attribute is **Money**. So, it is the next level node. Hence, out final decision tree is as:



2. Regression: Logistic Regression

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class.

- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.

Types of Logistic Regression:

Type	Description
Binomial Logistic Regression	Target variable has two categories (e.g., Yes/No)
Multinomial Logistic Regression	Target has 3+ unordered categories (e.g., cat/dog/sheep)
Ordinal Logistic Regression	Target has 3+ ordered categories (e.g., Low/Med/High)

Assumptions of Logistic Regression

- Independent Observations: Each sample is independent.
- Binary Output (for binary case): Target variable should be 0 or 1.
- Linearity of Logit: Linear relationship between predictors and log-odds.
- No Outliers: Outliers can distort parameter estimation.
- Sufficient Sample Size: Needed for stability and accuracy

Sigmoid Function

Sigmoid is a mathematical function that maps any real-valued number into a value between 0 and 1. Its characteristic "S"-shaped curve makes it particularly useful in scenarios where we need to convert outputs into probabilities. This function is often called the logistic function.

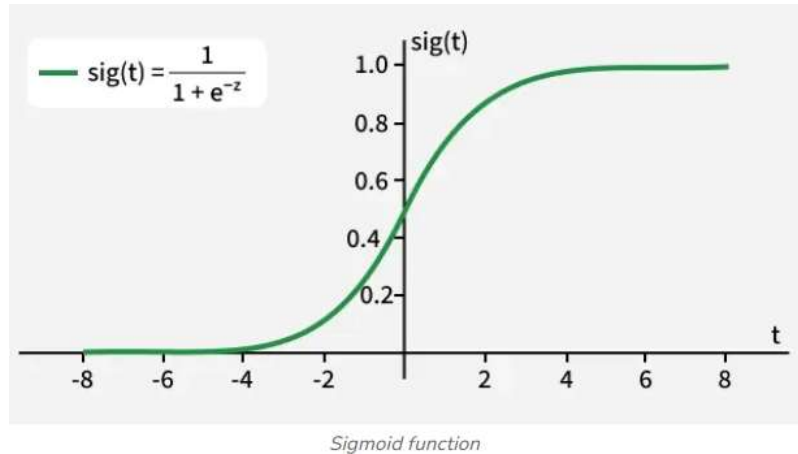
Mathematically, sigmoid is represented as:

$$\sigma = \frac{1}{1+e^{-x}}$$

where,

- x is the input value,
- e is Euler's number (≈ 2.718)

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As shown above the sigmoid function converts the continuous variable data into the probability i.e between 0 and 1.

- $\sigma(z)$ tends towards 1 as $z \rightarrow \infty$
- $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$
- $\sigma(z)$ is always bounded between 0 and 1

How it works

1. First, we take input features $X=[x_1, x_2, \dots, x_p]$
2. We compute a **linear combination** like in linear regression:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

3. But instead of directly predicting z , we apply the **sigmoid (logistic) function** to map z into a probability between 0 and 1:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Where:

- p = probability that the outcome = 1
 - $\sigma(z)$ = sigmoid function
4. We classify based on a threshold (usually 0.5):

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases}$$

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Applications

Logistic Regression is widely used in various domains due to its simplicity and interpretability. Some common applications include:

- Email Spam Detection
- Disease Prediction (e.g., Diabetes)
- Credit Risk Scoring
- Marketing Campaign Success Prediction

Advantages and Disadvantages

- **Advantages**
 - Simple and fast to implement.
 - Provides a probabilistic interpretation of the outcome.
 - Works well with linearly separable data.
- **Disadvantages**
 - Assumes a linear decision boundary, which may not hold for complex datasets.
 - May underperform when the relationship between features and target is nonlinear or intricate

Numerical:

A company wants to predict whether a student will get a job offer (1 = Yes, 0 = No) based on the number of programming projects completed. Assume $\beta_0 = -4, \beta_1 = 1.2$. The data collected is:

Projects Completed (X)	Job Offer (Y)
0	0
1	0
2	0
3	1
4	1
5	1

Task:

- a) Fit a logistic regression model:

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$$p = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 X$$

- b) Predict the probability of getting a job if a student completes 3 projects.
- c) Classify the outcome using a 0.5 threshold.

Solution:

Step 1: Given

$\beta_0 = -4$, $\beta_1 = 1.2$ and Classification threshold = 0.5

$$p = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 X$$

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases}$$

Step 2: Predict probability for $X = 3$

$$z = -4 + 1.2 \cdot 3 = -4 + 3.6 = -0.4$$

$$p = \frac{1}{1 + e^{-(-0.4)}} = \frac{1}{1 + e^{0.4}}$$

$$e^{0.4} \approx 1.4918$$

$$p = \frac{1}{1 + 1.4918} = \frac{1}{2.4918} \approx 0.401$$

Probability of getting a job ≈ 0.40 (40%)

Step 3: Optional – probabilities for all X

Projects (X)	$z = -4 + 1.2 X$	p (probability)
0	-4	0.018
1	-2.8	0.057
2	-1.6	0.167
3	-0.4	0.401
4	0.8	0.690
5	2	0.881

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- Predicted classes (0.5 threshold): 0,0,0,0,1,1
- Matches data trend reasonably.

Step 4: Final Answer:

- Probability of getting a job for 3 projects = **0.401 (40%)**
- Predicted class = **No job offer (0)**

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3. Variable Importance: Permutation tests, Partial dependence plots

In any predictive model (like Random Forest, Gradient Boosting, or even Logistic Regression), **some features (variables) are more influential** in making predictions than others.

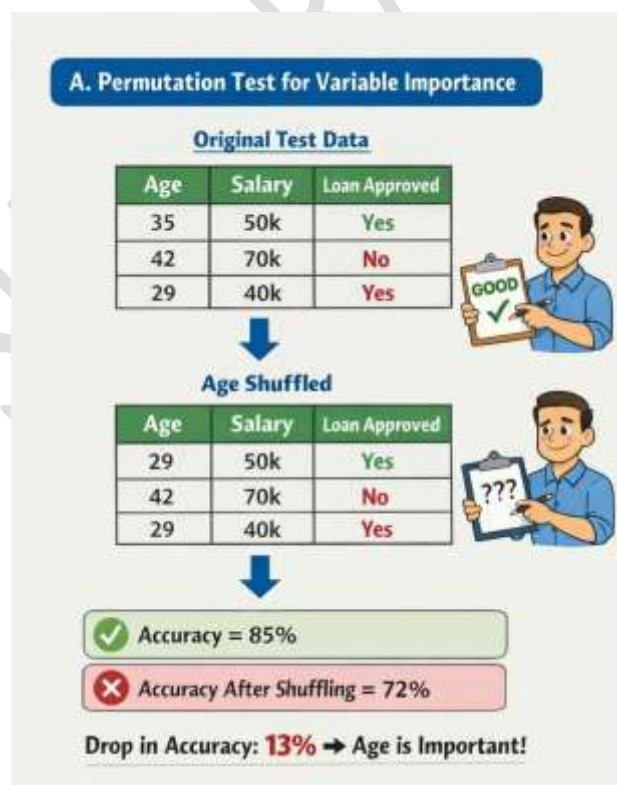
- **Variable importance** tells us **which features matter the most** for the model's predictions.
- Example: Predicting house prices:
 - Square footage → very important
 - Number of windows → less important

Knowing variable importance helps us:

- Understand the model
- Reduce unimportant features
- Make decisions based on key factors

a) Permutation Tests for Variable Importance

Idea: See how much the model's performance **drops** when a variable is “shuffled”.



Step by step:

1. Train your model normally.
2. Measure its **accuracy** (or RMSE, AUC, etc.) on a validation/test set.

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3. Pick one feature (e.g., “Age”) and **shuffle its values randomly**.
 - This breaks the relationship between that feature and the target.
4. Measure the model’s performance again.
5. Compute **importance score**:

$$\text{Importance} = \text{Original Performance} - \text{Shuffled Performance}$$

- **Big drop in performance** → **feature is important**
- **Small/no drop** → **feature is not important**

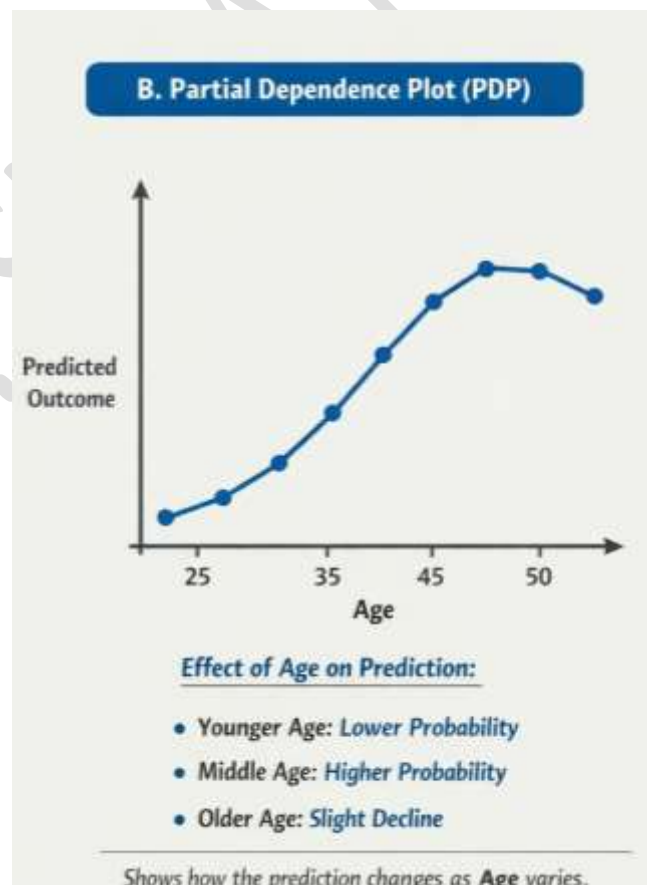
Analogy:

- Imagine you blindfold a chef and ask them to cook.
- Remove their ability to taste salt → dish quality drops a lot → salt is important.
- Remove the color of the plate → dish quality doesn’t change much → plate color not important.

b) Partial Dependence Plots (PDPs)

Idea: Show **how the prediction changes as one feature changes**, keeping all other features constant.

- Helps to **visualize the effect of a variable** on the predicted outcome.



4. Categorical and Mixed-type Data

Step by step:

1. Take a feature (e.g., Age).
2. For each possible value of Age:
 - Replace Age in all test examples with that value
 - Compute model predictions
 - Average the predictions across all examples
3. Plot **feature value vs average prediction** → PDP

Example: Predict house prices vs square footage:

Square footage	Predicted Price (Average)
500	100k
1000	200k
1500	300k

- The plot shows a **positive relationship**: more square footage → higher predicted price.

Key point: PDP shows **effect of a variable** while **marginalizing over all other variables**.

Summary

Concept	What it does	How it works
Variable Importance	Measures which features matter most	Based on model performance or contribution
Permutation Tests	Check importance by shuffling a feature	Drop in accuracy = importance score
Partial Dependence Plots	Visualize effect of one feature	Plot average predicted outcome vs feature value

Analogy together:

- Permutation test = “what happens if I remove this feature?”
- PDP = “if I change this feature, what happens to the prediction?”

4. Categorical and Mixed-type Data

4. Clustering: K-means and distance metrics for mixed-type data

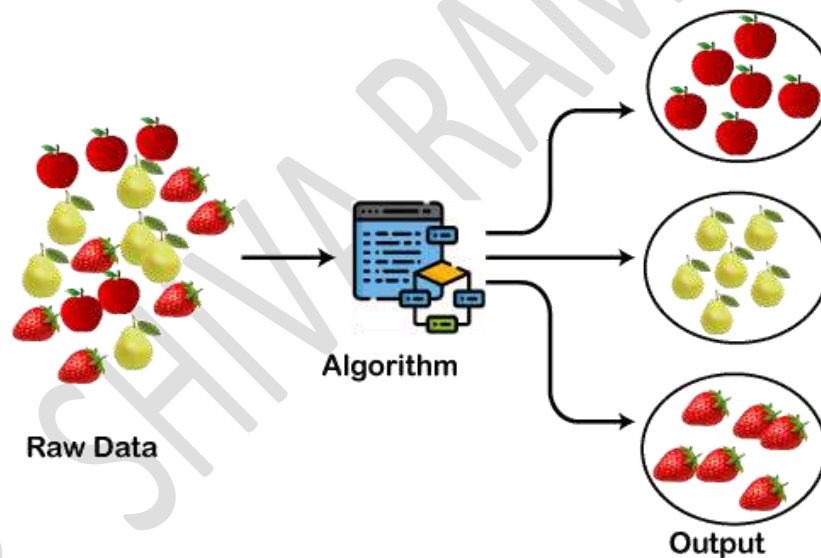
What is Clustering?

Clustering is an **unsupervised learning technique** where we group **similar data points together** based on their features.

In clustering techniques, **no label is given** for the model.

- Goal: Find **natural groups** in data without using labels.
- Example:
 - Customer segmentation: grouping customers by buying habits
 - Document clustering: grouping similar articles
 - Image clustering: grouping similar images

Key idea: Points in the **same cluster** are **more similar to each other** than to points in other clusters.



4. Categorical and Mixed-type Data

Application of Clustering:

Clustering analysis is broadly used in many applications such as market research, pattern recognition, data analysis, and image processing. Some of the major applications are:

- a) **Market Segmentation:** Identifying customer groups with similar purchasing behaviors.
- b) **Document Clustering:** Grouping similar text documents for organizing and summarizing information.
- c) **Biological Data Analysis:** Grouping genes or proteins with similar characteristics.
- d) **Anomaly Detection:** Identifying outliers in datasets.
- e) **Image Segmentation:** Dividing an image into meaningful parts for object detection or recognition.
- f) **Climate Analysis:** Clustering regions with similar weather patterns or environmental conditions.
- g) **Land Use Classification:** Grouping satellite images to classify areas as urban, agricultural, forest, etc.
- h) Any many more

K-means Clustering

- K-means is one of the **most popular clustering algorithms**.
- K-means algorithm is an **unsupervised learning algorithm**.
- Given a dataset of items, with certain features, and values for these features, the algorithm will categorize the items into K groups or clusters of similarity.
- To calculate the similarity, we can use the Euclidean distance, Manhattan distance, Hamming distance, Cosine distance as measurement.

K-means Algorithm:

1. **Choose K clusters** (number of groups).
2. **Randomly initialize K centroids** (center points of clusters).

4. Categorical and Mixed-type Data

3. **Assign each data point to the nearest centroid** (based on a distance metric, usually Euclidean).
4. **Recompute centroids** as the mean of points assigned to the cluster.
5. **Repeat steps 3–4** until the clusters don't change (convergence).

Mathematical Formulation

- Given dataset $X = \{x_1, x_2, \dots, x_n\}$
- Cluster assignment $C = \{C_1, C_2, \dots, C_K\}$
- Minimize **within-cluster sum of squares (WCSS)**:

$$\text{WCSS} = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

- Where:
 - μ_k = centroid of cluster k
 - $\|x_i - \mu_k\|$ = distance between point and centroid

Numerical 1:

Use K means algorithm to cluster the given datapoints where K=3: A(2, 10), B(2,5), C(8, 4), D(5, 8), E(7,5), F(6, 4), G(1, 2), H(4, 9)

Solution:

Suppose, initially we assign C1, C2 and C3 as the center of each cluster respectively. Let the initial centroids for cluster1, cluster2 and cluster 3 be **(2,10)** , **(5,8)** and **(1,2)** respectively.

Using Euclidian distance, we compute the distance of each data point with all the initial centroids. The data point having the minimum distance fall to that corresponding cluster.

4. Categorical and Mixed-type Data

First Iteration: Computation of Euclidian distance and assigning clusters

Initial Centroids:

C1: 2, 10)

C2: 5, 8)

C3: 1, 2)

Data Points			Distance to						Cluster
			2	10	5	8	1	2	
A	2	10	0.00		3.61		8.06		1
B	2	5	5.00		4.24		3.16		3
C	8	4	8.49		5.00		7.28		2
D	5	8	3.61		0.00		7.21		2
E	7	5	7.07		3.61		6.71		2
F	6	4	7.21		4.12		5.39		2
G	1	2	8.06		7.21		0.00		3
H	4	9	2.24		1.41		7.62		2

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

We need to compute new centroid for the first iteration.

New centroid calculation:

C1= (2, 10)

C2 = $(\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5}) = (6, 6)$

C3 = $(\frac{2+1}{2}, \frac{5+2}{2}) = (1.5, 3.5)$

Second Iteration: Computation of Euclidian distance and assigning new clusters

Current Centroids:

c1: (2, 10)

c2: (6, 6)

c3: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	3.5		
A	2	10	0.00		5.66		6.52		1	1
B	2	5	5.00		4.12		1.58		3	3
C	8	4	8.49		2.83		6.52		2	2
D	5	8	3.61		2.24		5.70		2	2
E	7	5	7.07		1.41		5.70		2	2
F	6	4	7.21		2.00		4.53		2	2
G	1	2	8.06		6.40		1.58		3	3
H	4	9	2.24		3.61		6.04		2	1

Since, data point **H** shifts from cluster 2 to cluster 1, we need to compute new centroids.

New centroid calculation:

C1= $(\frac{2+4}{2}, \frac{10+9}{2}) = (3, 9.5)$

C2= $(\frac{8+5+7+6}{4}, \frac{4+8+5+4}{4}) = (6.5, 5.25)$

C3= $(\frac{2+1}{2}, \frac{5+2}{2}) = (1.5, 3.5)$

4. Categorical and Mixed-type Data

Third Iteration: Computation of Euclidian distance and assigning new clusters

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A	2	10	1.12		6.54		6.52		1	1
B	2	5	4.61		4.51		1.58		3	3
C	8	4	7.43		1.95		6.52		2	2
D	5	8	2.50		3.13		5.70		2	1
E	7	5	6.02		0.56		5.70		2	2
F	6	4	6.26		1.35		4.53		2	2
G	1	2	7.76		6.39		1.58		3	3
H	1	2	7.76		6.39		1.58		3	3
I	4	9	1.12		4.51		6.04		1	1

Since, data point D shifts from cluster 2 to cluster 1, we need to compute new centroids.

New centroid calculation:

$$C1 = \left(\frac{2+5+4}{3}, \frac{10+8+9}{3} \right) = (3.67, 9)$$

$$C2 = \left(\frac{8+7+6}{3}, \frac{4+5+4}{3} \right) = (7, 4.33)$$

$$C3 = \left(\frac{2+1}{2}, \frac{5+2}{2} \right) = (1.5, 3.5)$$

Fourth Iteration: Computation of Euclidian distance and assigning clusters

Data Points			Distance to						Cluster	New Cluster
			3.67	9	7	4.33	1.5	3.5		
A	2	10	1.94		7.56		6.52		1	1
B	2	5	4.33		5.04		1.58		3	3
C	8	4	6.62		1.05		6.52		2	2
D	5	8	1.67		4.18		5.70		1	1
E	7	5	5.21		0.67		5.70		2	2
F	6	4	5.52		1.05		4.53		2	2
G	1	2	7.49		6.44		1.58		3	3
H	1	2	7.49		6.44		1.58		3	3
I	4	9	0.33		5.55		6.04		1	1

Since there is **no change in the clusters**, no need of New centroid calculation:

Hence the three clusters are:

Cluster 1: A(2, 10), H(4, 9), D(5, 8),

Cluster 2: C(8, 4), E(7, 5), F(6, 4)

Cluster 3: B(2,5), G(1, 2)

4. Categorical and Mixed-type Data

Numerical 2:

Generate cluster from following dataset using K-means algorithm (take k=2 and consider up to 2 iterations)

- T1: Bread, Jelly, Butter
- T2: Bread, Butter
- T3: Bread, Milk, Butter
- T4: Coke, Bread
- T5: Coke, Milk

Solution:

Convert Categorical Data to Binary Vectors using one-hot encoding

Transaction–Item Matrix

Transaction	Bread	Butter	Jelly	Milk	Coke
T1	1	1	1	0	0
T2	1	1	0	0	0
T3	1	1	0	1	0
T4	1	0	0	0	1
T5	0	0	0	1	1

Assume centroids as:

- C1 = T1
- C2 = T4

Centroid	Bread	Butter	Jelly	Milk	Coke
C1	1	1	1	0	0
C2	1	0	0	0	1

Iteration 1: Distance Calculation

Using Euclidean distance

$$d(X, Y) = \sqrt{\sum (x_i - y_i)^2}$$

4. Categorical and Mixed-type Data

Distance Table

Transaction	d to C1	d to C2	Cluster
T1	0	$\sqrt{3} = 1.73$	C1
T2	$\sqrt{1} = 1$	$\sqrt{2} = 1.41$	C1
T3	$\sqrt{2} = 1.41$	$\sqrt{3} = 1.73$	C1
T4	$\sqrt{3} = 1.73$	0	C2
T5	$\sqrt{4} = 2$	$\sqrt{2} = 1.41$	C2

Clusters formed are:

Cluster	Members
C1	T1, T2, T3
C2	T4, T5

Update Centroids

New Centroid C1= Mean of T1, T2, T3

New Centroid C2= Mean of T4, T5:

Cluster	Bread	Butter	Jelly	Milk	Coke	New Centroid
C1	$(1+1+1)/3 = 1$	$(1+1+1)/3 = 1$	$(1+0+0)/3 = 0.33$	$(0+0+1)/3 = 0.33$	0	C1 = (1, 1, 0.33, 0.33, 0)
C2	$(1+0)/2 = 0.5$	0	0	$(0+1)/2 = 0.5$	$(1+1)/2 = 1$	C2 = (0.5, 0, 0, 0.5, 1)

Iteration 2 – Distance Calculation

Transaction	d to C1	d to C2	Cluster
T1	0.47	1.58	C1
T2	0.47	1.32	C1
T3	0.94	1.32	C1
T4	1.15	0.71	C2
T5	1.63	0.71	C2

4. Categorical and Mixed-type Data

Convergence Check

Iteration	C1	C2
Iteration 1	T1,T2,T3	T4,T5
Iteration 2	T1,T2,T3	T4,T5

No change → Stop

Final Clusters

Cluster	Transactions	Interpretation
C1	T1, T2, T3	Bread–Butter dominant
C2	T4, T5	Coke–Milk dominant

Hamming Distance:

If you have two sequences X and Y of length n:

Hamming distance(X,Y) = Number of positions where $X_i \neq Y_i$

Example 1: Binary Vectors

- $X=[1,0,1,1,0]$
- $Y=[1,1,0,1,0]$

Compare position by position:

Position	X	Y	Different?
1	1	1	No
2	0	1	Yes
3	1	0	Yes
4	1	1	No
5	0	0	No

Hamming distance = 2 (positions 2 and 3 differ)

4. Categorical and Mixed-type Data

Example 2: Strings

- $X = "10101"$
- $Y = "11100"$

Compare each character: differences at positions 2, 3, 5 $\rightarrow$ Hamming distance = 3

Example 3: compute the hamming distance of all transactions with T1

- **T1: Bread, Jelly, Butter**
- **T2: Bread, Butter**
- **T3: Bread, Milk, Butter**
- **T4: Coke, Bread**
- **T5: Coke, Milk**

Step 1: Binary Representation

List of all items: Bread, Jelly, Butter, Milk, Coke

Transaction	Bread	Jelly	Butter	Milk	Coke
T1	1	1	1	0	0
T2	1	0	1	0	0
T3	1	0	1	1	0
T4	1	0	0	0	1
T5	0	0	0	1	1

Step 2: Compare each transaction with T1

T1 vs T1: [1,1,1,0,0] vs [1,1,1,0,0]

- Differences = 0, Hamming distance = 0

T1 vs T2: [1,1,1,0,0] vs [1,0,1,0,0]

- Differences = 1
Hamming distance = 1

T1 vs T3: [1,1,1,0,0] vs [1,0,1,1,0]

- Differences: Jelly (1 vs 0), Milk (0 vs 1) $\rightarrow$ 2 differences
Hamming distance = 2

4. Categorical and Mixed-type Data

T1 vs T4: [1,1,1,0,0] vs [1,0,0,0,1]

- Differences: Jelly (1 vs 0), Butter (1 vs 0), Coke (0 vs 1) → 3 differences

Hamming distance = 3

T1 vs T5: [1,1,1,0,0] vs [0,0,0,1,1]

- Differences: Bread, Jelly, Butter, Milk, Coke → 5 differences

Hamming distance = 5

Transaction	Hamming Distance from T1
T1	0
T2	1
T3	2
T4	3
T5	5

Assignment:

Generate cluster from following dataset using K-means algorithm (take k=2 and consider up to 2 iterations, use hamming distance).

- T1: Bread, Jelly, Butter**
- T2: Bread, Butter**
- T3: Bread, Milk, Butter**
- T4: Coke, Bread**
- T5: Coke, Milk**

4. Categorical and Mixed-type Data

Distance Metrics for Mixed-Type Data

K-means normally uses **Euclidean distance**, which works **well for numeric data**.

But many datasets have **mixed types**:

- Numeric (age, salary)
- Categorical (gender, country, product type)

K-means fails on mixed-type data because it relies on Euclidean distance and mean computation, which are not defined for categorical variables, leading to misleading distances and meaningless centroids.

Problem with Euclidean distance for categorical data

- Euclidean distance can't measure similarity between categories like "Red" vs "Blue".

Solution: Use special distance metrics for mixed data

1. Gower Distance

- Can handle **numeric + categorical + binary** features
- Scales numeric features between 0–1
- Categorical features: 0 if same, 1 if different
- Overall distance = weighted average of feature-wise distances

2. Hamming Distance (for categorical/binary features)

- Counts how many features are **different**

3. KModes or KPrototypes

- **K-modes**: clustering purely categorical data (uses modes instead of means)
- **K-prototypes**: handles **mixed data** (numeric + categorical)
 - Uses Euclidean for numeric, Hamming for categorical
 - Combines them with a weight factor

4. Categorical and Mixed-type Data

Numerical 1:

A company wants to cluster customers based on **Age (numeric)** and **Gender (categorical)**.

Use **K-prototypes clustering** with **K = 2** clusters. Assume:

- Weight factor $\gamma=1$
- Initial cluster prototypes:
 - **Cluster 1** $\rightarrow$ A (25, Male)
 - **Cluster 2** $\rightarrow$ C (40, Female)

Dataset

Customer	Age	Gender
A	25	Male
B	27	Male
C	40	Female
D	42	Female

Solution:

Distance formula (K-prototypes)

$$D(x, c) = (Age_x - Age_c)^2 + \gamma \cdot \delta(Gender_x, Gender_c)$$

Where:

- $\delta = 0$ if same category
- $\delta = 1$ if different category

Iteration 1: Distance Calculation Table and Cluster Formation

Customer	To C1 (25, Male)	To C2 (40, Female)	Assigned Cluster
A (25, M)	$(25-25)^2 + 0 = \mathbf{0}$	$(25-40)^2 + 1 = \mathbf{226}$	C1
B (27, M)	$(27-25)^2 + 0 = \mathbf{4}$	$(27-40)^2 + 1 = \mathbf{170}$	C1
C (40, F)	$(40-25)^2 + 1 = \mathbf{226}$	$(40-40)^2 + 0 = \mathbf{0}$	C2
D (42, F)	$(42-25)^2 + 1 = \mathbf{290}$	$(42-40)^2 + 0 = \mathbf{4}$	C2

4. Categorical and Mixed-type Data

Cluster Formation

Cluster	Members
C1	A, B
C2	C, D

Prototype Update Table

Cluster	Age (Mean)	Gender (Mode)	New Prototype
C1	$(25+27)/2 = 26$	Male	(26, Male)
C2	$(40+42)/2 = 41$	Female	(41, Female)

Iteration 2: Distance Calculation Table and Cluster Formation

Customer	To C1 (26,M)	To C2 (41,F)	Assigned
A (25,M)	$(-1)^2 + 0 = 1$	$(-16)^2 + 1 = 257$	C1
B (27,M)	$(1)^2 + 0 = 1$	$(-14)^2 + 1 = 197$	C1
C (40,F)	$(14)^2 + 1 = 197$	$(-1)^2 + 0 = 1$	C2
D (42,F)	$(16)^2 + 1 = 257$	$(1)^2 + 0 = 1$	C2

Final Clustering Result

Cluster	Prototype	Members
C1	(26, Male)	A, B
C2	(41, Female)	C, D

Key Points

Topic	Explanation
Clustering	Grouping similar data points without labels
K-means	Iterative algorithm minimizing distance to cluster centroids
Distance metric (numeric)	Euclidean distance
Distance metric (categorical)	Hamming or matching coefficient
Mixed-type data	Use Gower distance or K-prototypes
